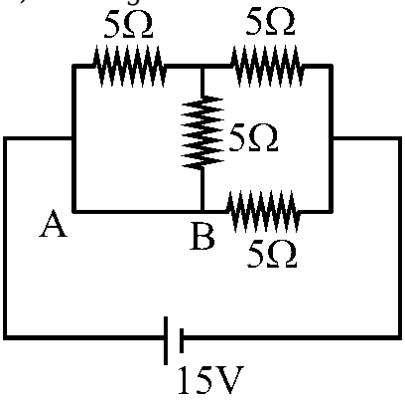
PART 1 : PHYSICS

SECTION-I (i)

1) In the given circuit find the magnitude of current flowing through arm 'AB' :



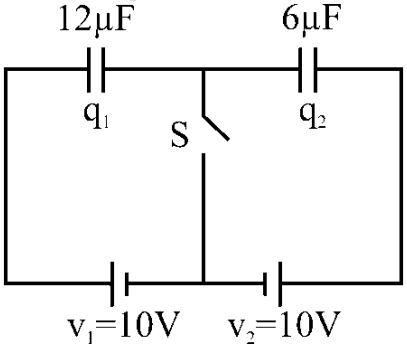
(A) 1.5A

(B) 2.5 A

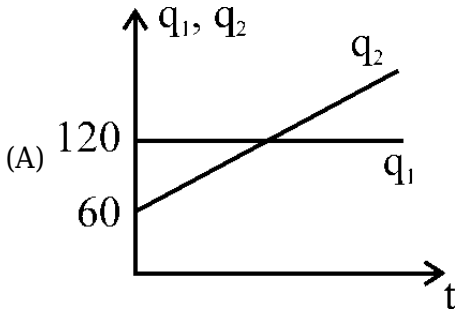
(C) 3.5A

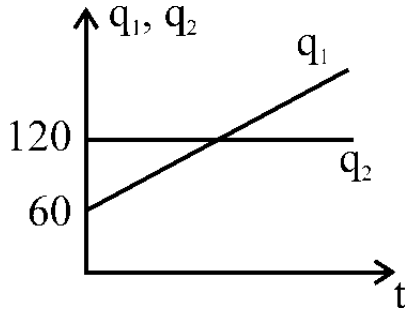
(D) 4.0A

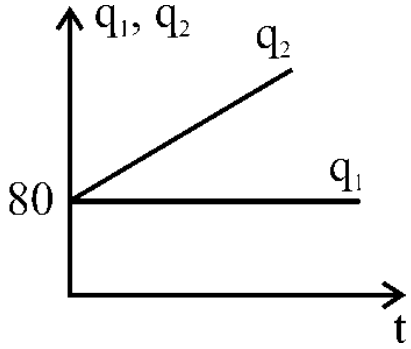
2) In the figure switch S is closed and steady state is reached. Now at t = 0 voltage V_2 is increased according to equation V = t + 10; upto the breakdown stage. The correct plot of charges q_1 and q_2 with respect to t will be :-

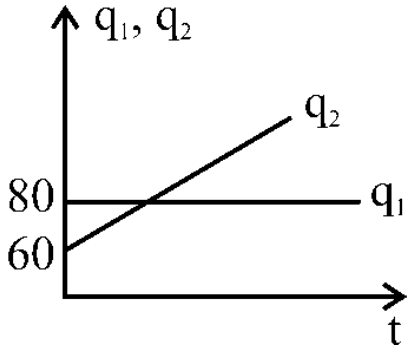


(A)



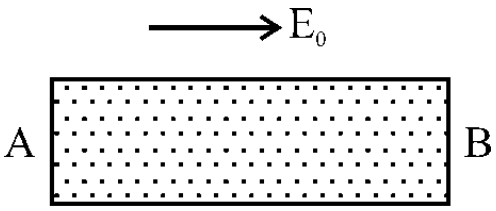
(B) 

(C) 

(D) 

3)

A metallic rod is placed in a uniform electric field. Select the **CORRECT** option.



(A) Inside the rod there will be an induced electric field from B to A which cancels external uniform field in the rod.

(B) Free electrons will accumulate at the end B of the rod.

(C) The potential of the end A will be more than that at B.

(D) The electric field outside the rod will not change due to the induced charges in the rod.

4) A projectile is projected with velocity kv_e in vertically upward direction from the ground into the space. (v_e is escape velocity and $k < 1$). If air resistance is considered to be negligible then the maximum distance from the centre of earth to which it can go, will be: (R = radius of earth)

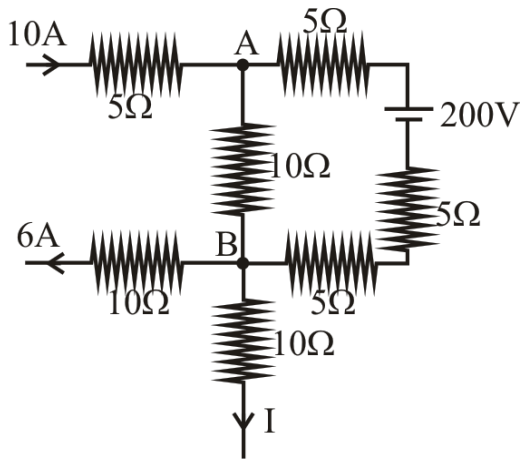
(A) $\frac{R}{k^2+1}$

(B) $\frac{R}{k^2-1}$

(C) $\frac{R}{1-k^2}$

(D) $\frac{R}{k+1}$

SECTION-I (ii)



1) In the given diagram choose correct options :-

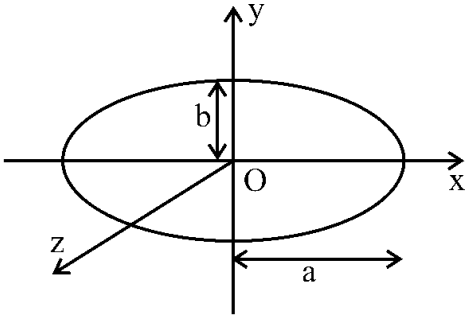
(A) The value of I is 4A

(B) The current from the cell is 4A.

(C) Voltage drop across AB is 140 V.

(D) The current from the cell is 10A.

2) An ellipsoidal gaussion surface with semi major axis a and semi-minor axis b is shown in figure.



An electric field $\vec{E} = \frac{C(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}}$ exist in space.

(A) Net flux through the gaussion surface is $4\pi C$

(B) Net charge enclosed by gaussion surface is $4\pi C\varepsilon_0$

(C) If given surface is a conductor electric potential at all its inside points may be constant for given field function

(D) If given surface is a conductor electric field at all its inside points may be constant for given field function

3) We have a parallel plate capacitor made of two conducting, circular sheets of radius 2R each. They are kept at a small separation d between them. Suppose, we insert a circular shape dielectric of radius R and thickness d, between the gap with its centre coinciding with the plate centres the dielectric constant is k. The correct statement is/are :

(A) Capacitance of the capacitor is $\frac{(k+3)\in_0 \pi R^2}{d}$

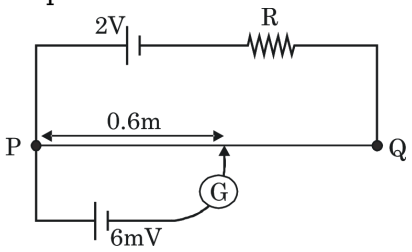
(B) Capacitance of the capacitor is $\frac{k\in_0 \pi \left(4R^2\right)}{d}$

(C) Ratio of electric field in the region with dielectric & without dielectric is 1

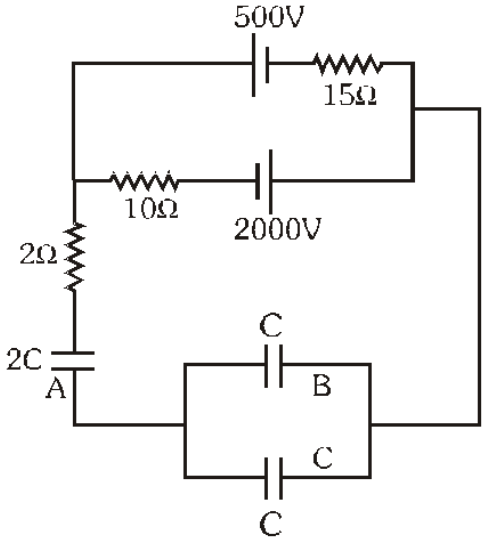
(D) Ratio of electric field in the region with dielectric & without dielectric is 1 : k

SECTION-II

1) In the shown potentiometer arrangement the wire PQ has a resistance of 5Ω and the driver cell has an emf of 2V. If a balance point is obtained at 0.6m along PQ, what is the value of resistance R (in Ω) ? (Length PQ = 1 m)

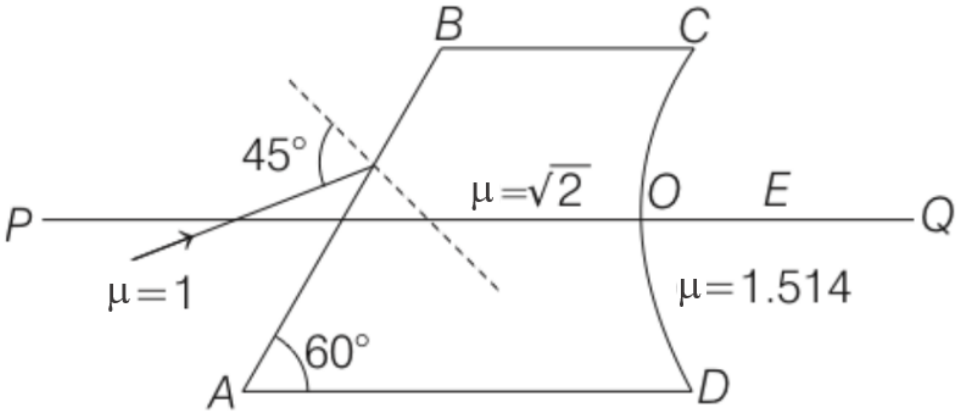


2) If C is equal to 2000 μF then find the summation of charges (in C) on all capacitors after steady state.



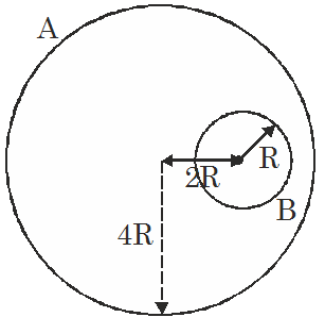
3) Two charged balls are connected by an inextensible thread of length 3 m. Masses of balls are 2 kg and 1 kg, the charges are +20 μC and –100 μC. What minimum constant external force F must be applied to the ball of mass 1 kg so that the thread does not slack? Neglect gravity and friction.

4) Figure shows an irregular block of material of refractive index $\sqrt{2}$. A ray of light strikes the face AB as shown in the figure. After refraction it is incident on a spherical surface CD of radius of curvature 0.4m and enters a medium of refractive index 1.514 to meet PQ at E. Find the distance OE upto two places of decimal.



5) Figure shows a uniformly charged spherical shell B of charge (q_B = q) kept inside uniformly charged spherical shell A of charge (q_A = q). Let electric field due to A and B at any point are $\vec{E}_A$ and $\vec{E}_B$ respectively. If dV represents elementry volume, then the value

$$\frac{\int_{\text{entire space}} \vec{E}_B . \vec{E}_B dV}{\int_{\text{entire space}} \vec{E}_A . \vec{E}_B dV}$$



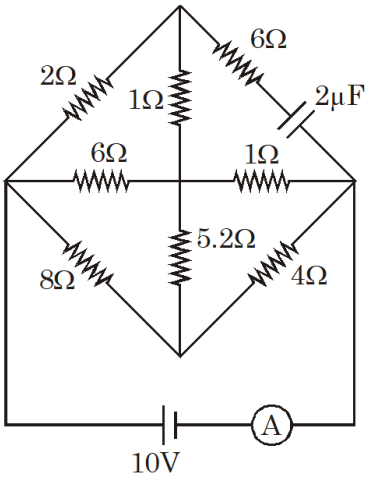
of will be

6)

A thin wire is wound very tightly in one layer on the surface of a sphere of paramagnetic material (μ_r = 1). The planes of all the turns can be assumed to be perpendicular to the same diameter of the sphere. The turns cover the entire surface of the sphere. The radius of the sphere is R, the total number of turns is N, and the current in the winding is I. Find the value of magnetic induction at the centre of the sphere. (Given : μ_0NI = 20 R, assume that width of the wire is constant)

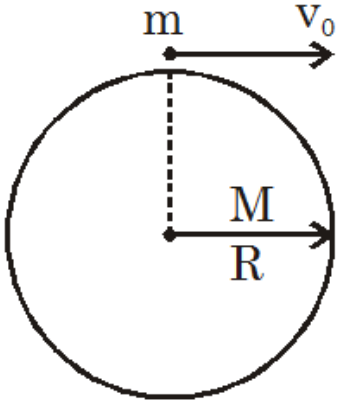
7)

In the given circuit find reading of ammeter (in A) when capacitor has attained steady state.



8) Given that mass of Earth is M and that of moon is m and distance between them is R. Neglect effect of any other planet or sun. A small mass is projected from midway from Earth and moon its escape velocity is $P\sqrt{\frac{G}{R}(m+M)}$ find P.

9) A body of mass m is projected horizontally just above the earth's surface with velocity $v_0 = \sqrt{\frac{4}{3}gR}$

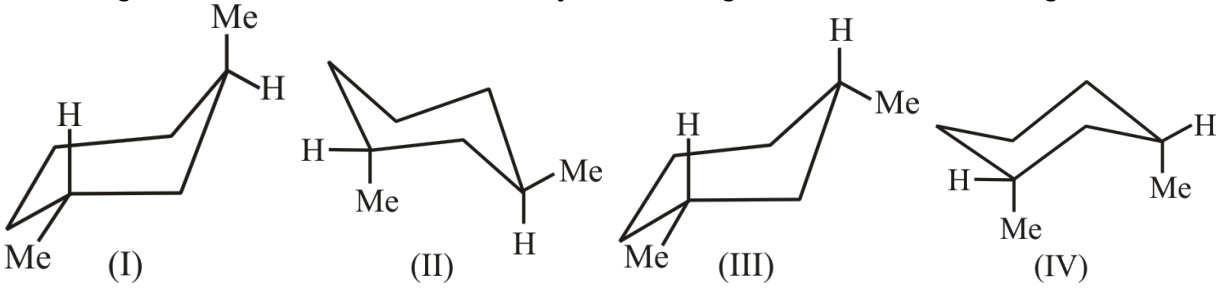


. The maximum height attained by the body is $\frac{2R}{3n}$. Find n.

PART 2 : CHEMISTRY

SECTION-I (i)

1) Arrange in the correct order of stability (decreasing order) for the following molecules



(A) (I) > (II) > (III) > (IV)

(B) (IV) > (III) > (II) » (I)

(C) (III) > (I) = (II) > (IV)

(D) (I) > (ll) = (III) > (IV)

2) At 300 K a first order reaction is 50% completed in 20 minutes. At 350K the same reaction is 50% completed in 5 minutes in similar conditions. Activation energy of the reaction is nearly : [Given : ln2 = 0.7]

(A) 2.88 kcal/mol

(B) 12.88 kcal/mol

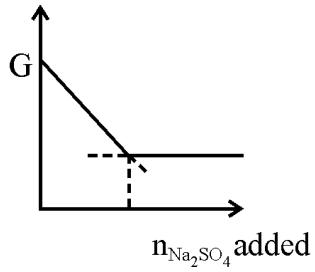
(C) 8.88 kcal/mol

(D) 5.88 kcal/mol

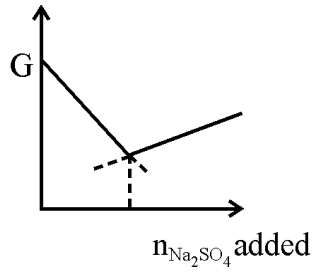
3) Which of the following closely represents the variation of conductance of a dilute $BaCl_2$ solution which is gradually contaminated by slowly adding Na_2SO_4 ?

(Given limiting molar conductances : $\Lambda^0_{Na^+}$ = 50.11 Scm^2 / mole, $\Lambda^0_{Ba^{2+}}$ = 127 Scm^2 / mole).
[Plot given is between conductance of solution (G) vs. moles of Na_2SO_4 added]

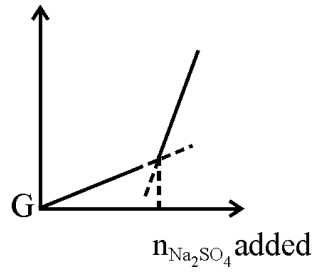
(A)



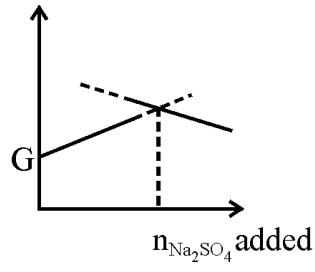
(B)



(C)



(D)



4)

The correct order of energy of absorption for the following metal complexes is -
A : $[Ni(en)_3]^{2+}$, B : $[Ni(NH_3)_6]^{2+}$, C : $[Ni(H_2O)_6]^{2+}$

(A) C < B < A

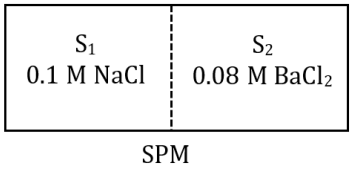
(B) B < C < A

(C) C < A < B

(D) A < C < B

SECTION-I (ii)

1) Two solutions S_1 and S_2 containing 0.1M NaCl(aq.) and 0.08M $BaCl_2$(aq.) are separated by semipermeable membrane. Which among the following statement(s) is/are correct -



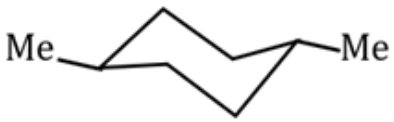
(A) S_1 and S_2 are isotonic

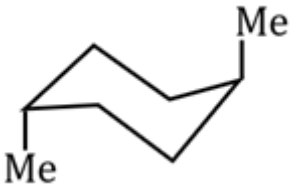
(B) S_1 is hypertonic and S_2 is hypotonic

(C) S_1 is hypotonic and S_2 is hypertonic

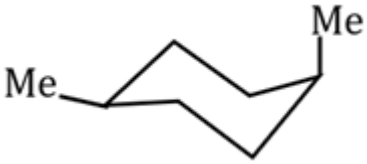
(D) Osmosis will take place to from S_1 to S_2

2) Select the correct statement for the following species.

(I) 

(II) 

(III) 

(IV) 

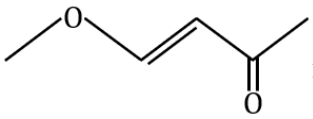
(A) I & II are geometrical isomers

(B) II & III are geometrical isomers

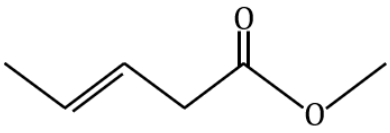
(C) III & IV are conformers

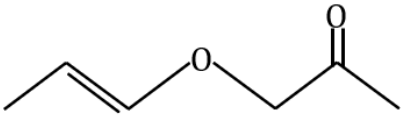
(D) I & IV are conformers

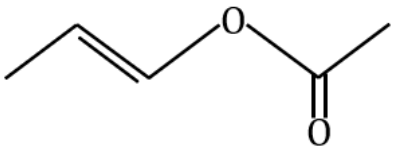
3) Compound(s) having higher bond rotation energy barrier across C = C than compound.

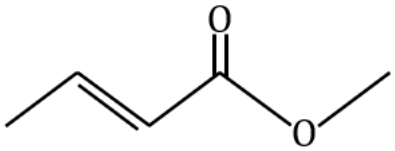


is/are-

(A) 

(B) 

(C) 

(D) 

SECTION-II

1) The ionization potential of hydrogen atom is 13.6 eV. The energy in electron volt required to remove an electron in the n = 2 state of the hydrogen atom is

2) A sample of 125 ml H_2O_2 (aq) solution required 200 ml 0.1M $KMnO_4$ for the complete reaction. Calculate volume strength of H_2O_2 at STP (1bar & 0°C.)
$2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5O_2$

3)

Calculate molality of a solution obtained by mixing equal volume of 30% w/w (density = 1.2gm/ml) & 70 % w/w (density = 1.6 gm/ml) of solute. (Molecular weight of solute is 100.)

4)

Assume isotope of chlorine present on the unknown planet are ^{34}Cl and ^{38}Cl. If average molecular weight of Cl is found to be 35. What is the sum of moles of proton and neutron in 7 gm of sample of chlorine

5)

A certain reactant A^{n+} is getting converted to $A^{(n+4)+}$ in solution. The rate constant of this reaction is measured by titrating a volume of the solution with a reducing reagent which only reacts with A^{n+} and $A^{(n+4)+}$. In this process, it converts A^{n+} to $A^{(n-2)+}$ and $A^{(n+4)+}$ to $A^{(n-1)+}$.

Time	0	10 min
Volume of reagent consumed	30 ml	45 ml

Calculate the rate constant (in min^{-1}) of the conversion of A^{n+} to $A^{(n+4)+}$ assuming it to be first order reaction.
[n-factor of reagent remain same when it reacts with A^{n+} & $A^{(n+4)+}$]
[Given : ln 3/2 = 0.4]

6) Lead storage battery is used in automobiles. In order to reduce the amount of hazardous lead and compounds that end up in landfills when these batteries are dumped, it has been suggested that the lead be replaced with carbon monoxide. The unbalanced cell reaction when the battery discharge is :
$PbO_2(s) + CO(g) + H_2SO_4(aq) \longrightarrow PbSO_4(s) + CO_2(g) + H_2O$
If density of sulphuric acid in battery decreases from 1.3 g/ml (40% H_2SO_4 by wt.) to 1.2 g/ml (20% H_2SO_4 by wt.) during its used, determine the ampere hour for which battery has been used. The constant volume of electrolytic solution is 3.5 L.

7) Number of molecules are having zero dipole moment are X.
Number of molecules which are planar are Y
Express value of X + Y
PCl_3F_2, $P(CH_3)_3(CF_3)_2$, BF_3, $CHCl_3$, $P(CH_3)_2(CF_3)_3$, PF_3Cl_2, CH_4

8) If Z_{eff} (as per Slater's rule) of 3s electron in Fe is 'x' then find out the value of $\frac{x}{10}$.

9)

A potential difference of 20 kV is applied across an X-ray tube. Find the minimum wavelength (in Å) of X-ray generated.

PART 3 : MATHEMATICS

SECTION-I (i)

1) Consider a variable matrix $V = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$ such that $VA = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ ..... (E), where $A = \begin{bmatrix} \log a \\ \log b \\ \log c \end{bmatrix}$ and a, b, c are distinct positive reals in G.P.. If solutions (x, y, z) of matrix equation (E) lie on line $\frac{x}{\alpha} = \frac{y}{\beta} = \frac{z}{\gamma}$, then $\frac{\alpha}{\beta} + \frac{\beta}{\gamma} + \frac{\gamma}{\alpha}$ is (x, y, z ≠ 0)

(A) 2
(B) 4
(C) 3
(D) 5

2) Let A_1 is a matrix formed by replacing all the elements of a square matrix A of order n by its corresponding cofactors, A_2 is matrix formed by replacing all elements of matrix A_1 by its corresponding cofactors and A_3, A_4 ... formed so on. If $|A_n| = \lambda$, then | A | equals (where n > 1 and |A| represent determinant value of matrix A)

(A) λ^{n^n}
(B) $\lambda^{(n-1)^n}$

(C) $\lambda^{\frac{1}{(n-1)^n}}$

(D) $\frac{1}{\lambda^{(n-1)^n}}$

3) If $D_1 = \begin{vmatrix} ab^2-ac^2 & bc^2-a^2b & a^2c-b^2c \\ ac-ab & ab-bc & bc-ac \\ c-b & a-c & b-a \end{vmatrix}_7$ $D_2 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix}$, then $D_1\ D_2$ is equal to -

(A) 0

(B) D_1^2

(C) D_2^2

(D) D_2^3

4) If f(x) and g(x) are derivative and antiderivative of a function h(x) where $g\left(\frac{\pi}{2}\right) = 3$ and $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (f(x)+g(x)) \cos x\, dx = 5$. Let p(x) be a polynomial of degree 4 such that p(1) = 7 and attains its local minimum value 3 at both x = 2 and x = 3. If the local maximum value of p(x) is equal to λ then the value of $\lambda - g\left(\frac{-\pi}{2}\right)$ is

(A) $\frac{19}{8}$

(B) $\frac{17}{16}$

(C) $\frac{19}{16}$

(D) None of these

SECTION-I (ii)

1) 14 identical black balls and 14 identical white balls are to be arranged in a row. The number of ways to arrange them such that on picking any 7 consecutive balls we get atleast one black ball is ${}^{28}C_k - \ell.{}^{21}C_7 + {}^{15}C_m$, then

(A) $k + \ell$ can be 29

(B) k - m can be 12

(C) k = 19

(D) m, k, ℓ can be three consecutive term of an A.P.

2) The value(s) of the real parameter 'a' for which the inequality $x^6 - 6x^5 + 12x^4 + ax^3 + 12x^2 - 6x + 1 \geq 0$ is satisfied for all real x

(A) [–12, 37]

(B) [–12, 38]

(C) [–17, 14]

(D) [–17, 35]

3) Tangents are drawn from $\left(-\frac{\pi}{2}, 0\right)$ to f : [0, ∞) → R ; y = sinx and $(a_i, \sin a_i)$ are the points of contact of these tangents for i = 1, 2, 3, ... and $a_{i+1} > a_i$. Choose the **correct** options(s)

(A) $a_n + \frac{\pi}{2} = \tan a_n$

(B) $a_{n+1} - a_n < \pi$

(C) $a_{n+2} - a_n > 2\pi$

(D) $\sin a_{2n} - \sin a_{2n-1} < 0$

SECTION-II

1) Consider the following functions $f(x) = \tan\left(\pi x^3\right)\sqrt{9+\tan^2\pi x}\sqrt[3]{27+\tan^3\pi x}$ and $g(x) = \frac{\cos^{-1}x + \tan^{-1}x - \cos x\cos^{-1}x - \cos x\tan^{-1}x}{(1+x^2)\cot^{-1}x}$. The value of $(f'(-2) + \pi g''(0))$ is equal to kπ. Then k is equal to

2) If sum of the values of x satisfying the equation $\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)\left(\sec^2 x - (1+\sqrt{3})\tan x + 2 + \sqrt{3}\right) = \sqrt{9+18\sin x\cos x}$ in [0, 2π] is $\left(\frac{p}{q}\right)\pi$, where p and q are co-prime, then $|p-q|$ is equal to

3) $\prod_{k=1}^{2020}\left\{\left(1+2\cos\left(\frac{2\pi.3^k}{3^{2020}+1}\right)\right)\right\}$ is equal to

4) Suppose that f and g are non–constant continuous and differentiable, real valued functions on R. If for every x, y ∈ R, f(x + y) = f(x).f(y) – g(x).g(y) and g(x + y) = g(x).f(y) + f(x).g(y). Also f ′(0) = 0 then find the value of $\frac{1-\left\{(f(x))^6 + (g(x))^6\right\}}{(f(x))^2(g(x))^2}$.

5) Let $S = {}^{31}C_1 - \left(1+\frac{1}{2}\right){}^{31}C_2 + \left(1+\frac{1}{2}+\frac{1}{3}\right){}^{31}C_3 - \left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right){}^{31}C_4 + ... + \left(1+\frac{1}{2}+\frac{1}{3}+...+\frac{1}{31}\right){}^{31}C_{31}$, then $\frac{1}{S}$ is equal to

6) Let $\ell_1, \ell_2, ..., \ell_{100}$ be consecutive terms of an arithmetic progression with common difference d_1, and let $\omega_1, \omega_2, ..., \omega_{100}$ be consecutive terms of another arithmetic progression with common difference d_2,

where $d_1d_2 = 10$. For each i = 1, 2,...,100, let R_i be a rectangle with length ℓ_i, width ω_i and area A_i. If $A_{51} - A_{50} = 1000$, then the value of $A_{100} - A_{90}$ is

7) If the $\lim_{n\to\infty} \sum_{k=2}^{n} \cos^{-1}\left(\frac{1+\sqrt{(k-1)k(k+1)(k+2)}}{k(k+1)}\right) = \frac{120\pi}{k}$ then find the value of k.

8) If $\alpha, \beta;\ \beta, \gamma$ and γ, α are the roots of $a_ix^2 + b_ix + c_i = 0$ respectively for i = 1, 2, 3 (given that $\alpha, \beta, \gamma > 0$) respectively if $\sum\alpha + \sum\alpha\beta + \alpha\beta\gamma = \left(\prod_{i=1}^{3}\left(\frac{a_i - b_i + c_i}{a_i}\right)\right)^{\frac{1}{2}} + k$, then $|k|$ is equal to _____.

9) If $\log_8^4(x+1) + \log_3^2(\log_5 y) + \log_5^6(3 - \log_2 z) = 0$, then the number of zeroes after decimal before a significant digit occurs in $(x + yz)^{-100}$, is (where $\log_{10}2 = 0.301$)

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	D	A	A	C

SECTION-I (ii)

Q.	5	6	7
A.	A,B,C	A,B	A,C

SECTION-II

Q.	8	9	10	11	12	13	14	15	16
A.	995.00	4.00	3.00	6.05 to 6.06	4.00	5.00	4.16 to 4.17	2.00	0.67

PART 2 : CHEMISTRY

SECTION-I (i)

Q.	17	18	19	20
A.	C	D	B	A

SECTION-I (ii)

Q.	21	22	23
A.	C,D	B,C	A,B,C,D

SECTION-II

Q.	24	25	26	27	28	29	30	31	32
A.	3.38 to 3.43	4.50 to 4.60	11.10 to 11.25	7.00	0.04	536.00 to 536.20	5.00	1.46 to 1.48	0.62

PART 3 : MATHEMATICS

SECTION-I (i)

Q.	33	34	35	36
A.	C	C	D	B

SECTION-I (ii)

Q.	37	38	39
A.	A,B,D	B	A,C,D

SECTION-II

Q.	40	41	42	43	44	45	46	47	48
A.	**109.00**	**25.00**	**1.00**	**3.00**	**31.00**	**18900.00**	**720.00**	**1.00**	**130.00**

SOLUTIONS

PART 1 : PHYSICS

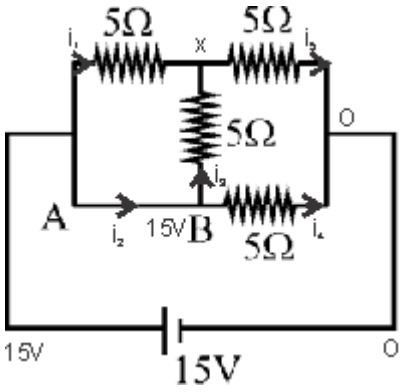
1)

Question Explanation :

We have to calculate the magnitude of current flowing through branch AB of given circuit.

Concept :
This question is based on Circuit Analysis.

Solution :



at node 'x'

$i_3 + i_1 = i_5$

$$\frac{15-x}{5} + \frac{15-x}{5} = \frac{x-0}{5} \qquad (i_1 = 1A,\ i_5 = 2A,\ i_4 = 3A,\ i_3 = 1A)$$

$15 + 15 = 3x$

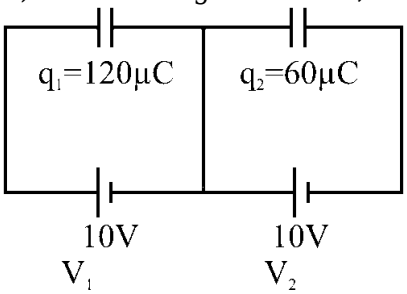
$x = 10\ V$

$i_2 = i_3 + i_4 = 1 + 3 + 4A$

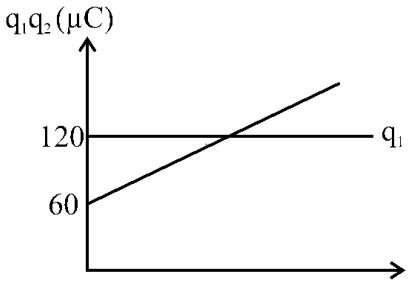
Current in arm AB is 4 A.

Hence,
Option (4) is correct

2) After closing the switch, the final charges on both capacitors are as shown.



Now when V_2 starts incraesing, then only q_2 starts increasing, but q_1 remains unaffected.



$\Rightarrow q_2 = C_2V_2$

$q_2 = 6\,(t + 10)$

3)

Conceptual

4) Kinetic energy = Potential energy

$$\frac{1}{2}m(ku_e)^2 = \frac{mgh}{1+\frac{h}{R}} \Rightarrow \frac{1}{2}mk^2\,2gR = \frac{mgh}{1+\frac{h}{R}} \Rightarrow h = \frac{Rk^2}{1-k^2}$$

Height of Projectile from the earth's surface = h

Height from the centre $r = R + h = R + \frac{Rk^2}{1-k^2}$

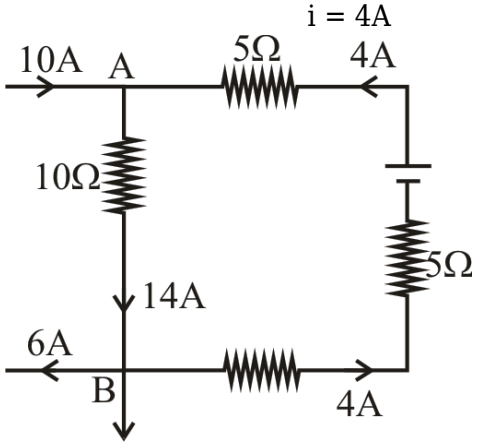
By solving $r = \frac{R}{1-k^2}$

5)

By KCL, $6 + I = 10 \Rightarrow I = 4A$

By KVL, $200 - 25\,i - 100 = 0$

$i = 4A$



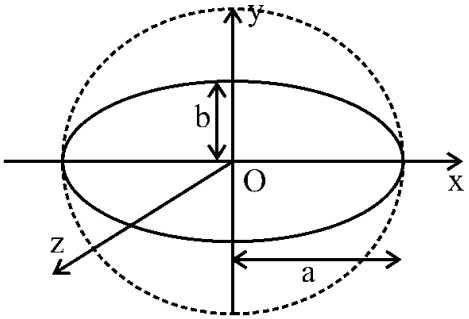
$V_{AB} = 10 \times 14 = 140\ V$

6) $\vec{E} = \frac{C\vec{r}}{r^3}$, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

which is electric field due to point charge.

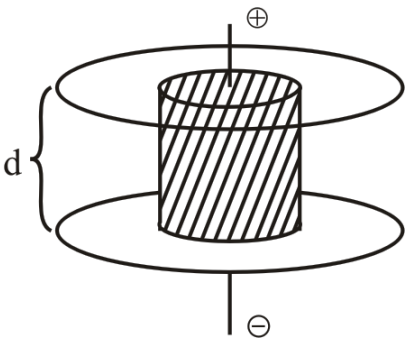
Flux through ellipse = flux through sphere of radius a

$$\phi = \vec{E} \cdot \vec{A} = \frac{C}{a^2} \times 4\pi a^2$$



Charge in closed = $\phi\varepsilon_0 = 4\pi C\ \varepsilon_0$

Electric field & potential will be variable.



7) $C = C_{di} + C_{empty}$

$$= \frac{k\in_0 \left(\pi R^2\right)}{d} + \frac{\in_0 \pi \left(3R^2\right)}{d}$$

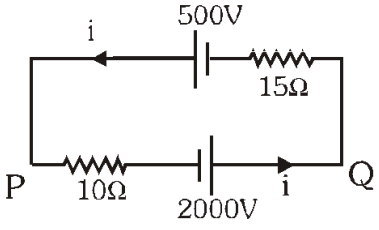
$$C = \frac{(k+3)\in_0 \pi R^2}{d}$$

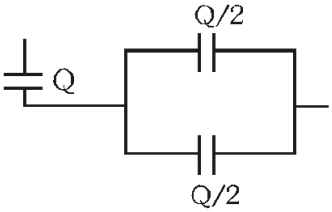
8)

Using basics concept of potentiometer.

9) 500 – 10 i + 2000 – 15 i = 0

$$\Rightarrow i = \frac{2500}{25} = 100A$$

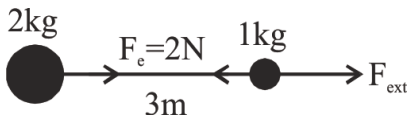




Now $V_{PQ} = 1000$ and $C_{eq} = C$

$\Rightarrow Q = V_{PQ} \times C = 2$

summation of charges = 2Q = 4

10) 

$F_{ext} = 3a \Rightarrow a = F_{ext} / 3$

$$F_e = \frac{9 \times 10^9 \times 2 \times 10^{-9}}{9} = 2(a) = \frac{2F_{ext}}{3}$$

$$\frac{2F_{ext}}{3} = 2 \Rightarrow F_{ext} = 3$$

11) Applying Snell's law on face AB

$$(1)\ \sin 45\gamma = \left(\sqrt{2}\right) \sin r$$

$$\therefore \sin r = \frac{1}{2}$$

or $r = 30\gamma$

i.e. ray becomes parallel to AD inside the block.

Now applying, $\frac{\alpha_2}{v} - \frac{\alpha_1}{u} = \frac{\alpha_2 - \alpha_1}{R}$ on face CD,

$$\frac{1.514}{OE} - \frac{\sqrt{2}}{\infty} = \frac{1.514 - \sqrt{2}}{0.4}$$

Solving this equation we get OE = 6.06 m

12) $$\int \vec{E}_B \cdot \vec{E}_B dV = \frac{2}{\epsilon_0} \int \frac{1}{2} \epsilon_0 \vec{E}_B \cdot \vec{E}_B dV$$

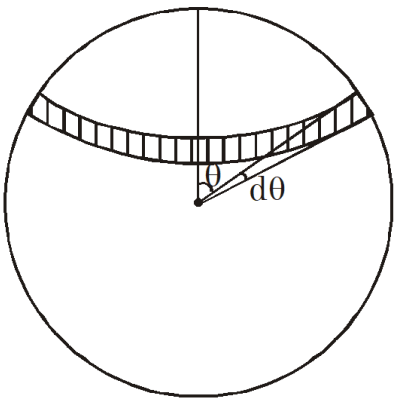
$$= \frac{2}{\epsilon_0} [\text{self energy of } q_B]$$

$$= \frac{2}{\epsilon_0} \left[\frac{1}{2} \frac{kq^2}{R}\right] = \frac{1}{\epsilon_0} \frac{kq^2}{R}$$

$$\int \vec{E}_A \cdot \vec{E}_B dV = \frac{1}{\epsilon_0} \int \epsilon_0 \vec{E}_A \cdot \vec{E}_B dV$$

$$= \frac{1}{\epsilon_0} [\text{Interaction energy of } q_A \text{ and } q_B]$$

$$= \frac{1}{\epsilon_0} \left[\frac{kq^2}{4R}\right]$$



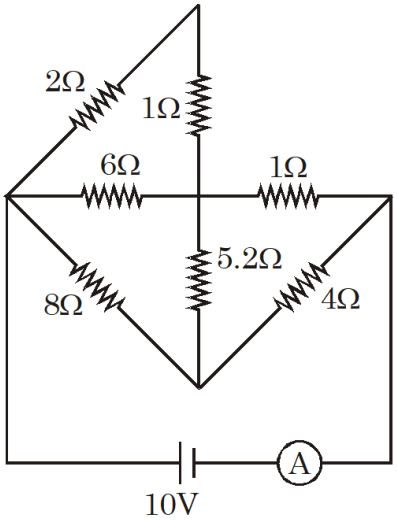
13)

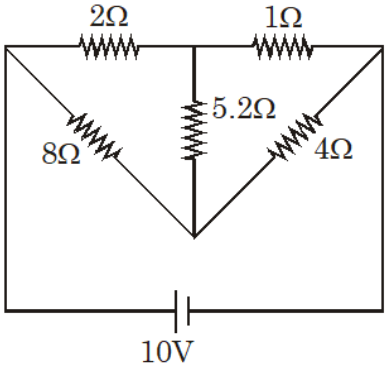
$$dN = \frac{N}{\pi R_\pi} R d\theta = \frac{N d\theta}{\pi}$$

$$\int dB = \int_0 \frac{\mu_0 i R^2 \sin^2\theta}{2R^3} \frac{N d\theta}{\pi}$$

$$= \frac{10}{\pi}\left[\frac{\theta - \frac{\sin 2\theta}{2}}{2}\right]_0^{\pi} = 5$$

14)



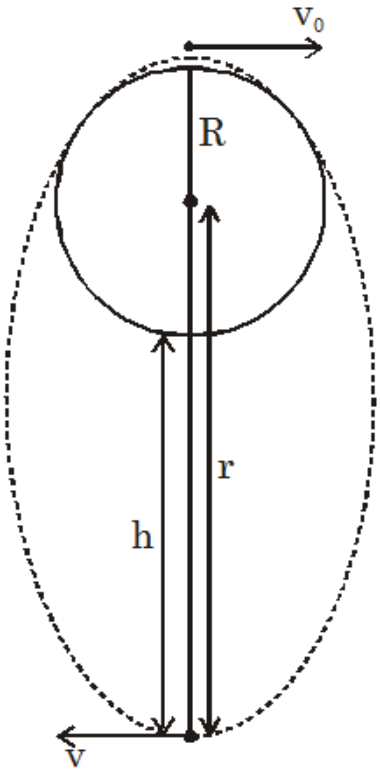


Its balanced wheatstone bridge

15) $|PE| = KE$

$$\left|\left(-\frac{Gm}{R/2} - \frac{GM}{R/2}\right)\right| M_1 = \frac{1}{2} M_1 V^2$$

$$\sqrt{\frac{4G}{R}(m+M)} = v \Rightarrow P = 2$$



16)

$L = mv_0R = mvr$

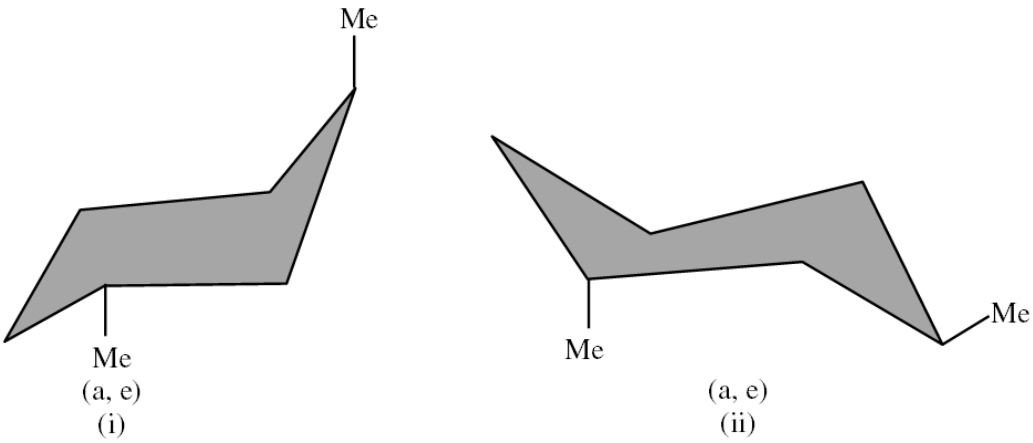
$$E = \frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

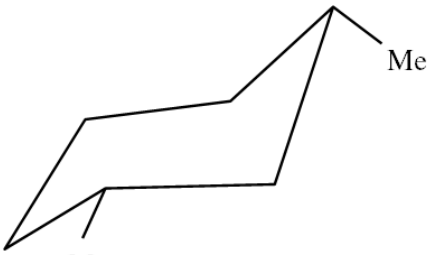
on solving, h = R

So $\frac{2}{3n} = 1 \Rightarrow n = \frac{2}{3} = 0.67$

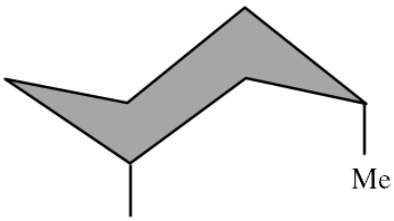
PART 2 : CHEMISTRY

17)





(e, e)
(iii)



(a, a)
(iv)

stb. order $\boxed{ee > ae/ea > aa}$
(iii) > (i) = (ii) > (iv)

18) $\ln \frac{\left(\frac{0.7}{5}\right)}{\left(\frac{0.7}{20}\right)} = \frac{E_a}{2 \times 10^{-3}}\left[\frac{1}{300} - \frac{1}{350}\right]$

$\frac{1.4 \times 2 \times 10^{-3} \times 300 \times 350}{50} = E_a$

E_a = 5.88 kcal/mol

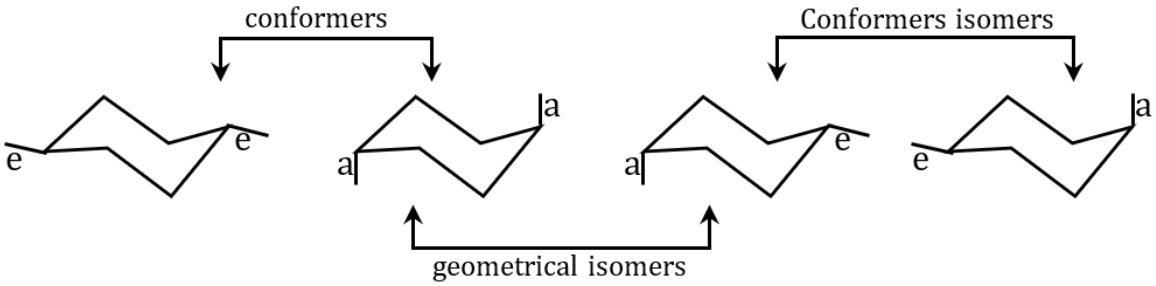
19) Conductance first decreases because $2\Lambda_{m_{Na^+}} < \Lambda_{m_{Ba^{2+}}}$ and then increases due to increase in total number of ions.

20)

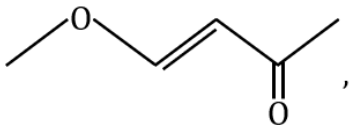
Stronger the ligand, larger the splittig & higher the energy of absorption.
$[Ni(en)_3]^{2+} > [Ni(NH_3)_6]^{2+} > [Ni(H_2O)_6]^{2+}$

21)

For solution S_1;
$\pi_1 = C_1RT \times i$; C_1 = 0.1 M
$i = 1 + (n - 1)\alpha$;
$NaCl \rightarrow Na^{\oplus} + Cl^{\ominus}$; n = 2; α = 1 (assuming 100% dissociation)
$\Rightarrow$ i = 2
$\Rightarrow$ $\pi_1 = 0.1 \times RT \times 2 = 0.2RT$
for solution (2);
$BaCl_2 \rightarrow Ba^{2\oplus} + 2Cl^{\ominus}$; n = 3; α = 1 (assuming 100% dissociation)
i = 3
$\Rightarrow$ $\pi_2 = (0.08)RT \times 3 = 0.24RT$
$\pi_2 > \pi_1 \Rightarrow S_2$ is hypertonic as compared to S_1



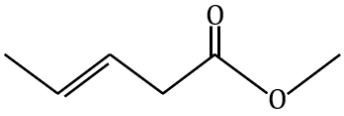
22)

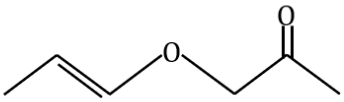
In  ,

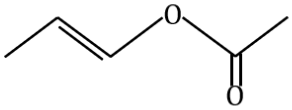
23) there is extended conugation.

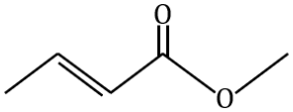
⇒ Bond rotation energy barrier order :

no conjugation > cross conjugation > extended conjugation

(A)  ⇒ no conjugation with C = C → higher

(B)  ⇒ cross conjugation → higher

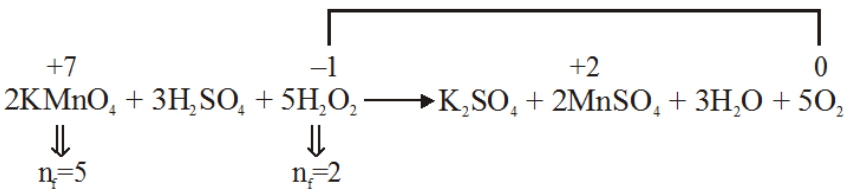
(C)  ⇒ cross conjugation → higher

(D)  ⇒ cross conjugation → higher

24)

$I.P = R_H Z^2$, $13.6 = R_H$

$$\Delta E = R_H\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right) = 13.6\left(\frac{1}{2^2}-\frac{1}{\infty^2}\right) = 13.6\times\frac{1}{4} = 3.4$$



25)

At equivalence point ;

$\boxed{\text{meq. of } H_2O_2 = \text{m eq. of } KMnO_4}$

milliequivalent = $M \times V_{mL} \times n_f$

$M \times 125 \times 2 = 0.1 \times 200 \times 5$

$\boxed{M_{H_2O_2} = 0.4\ M}$

Volume strength = $M \times 11.35$ at 1 bar, 0°C

$= 0.4 \times 11.35$

= 4.54 V of H_2O_2

Ans. 4.54

26) Suppose volume of solution in each case is 100 ml in first case weight of solute

$= \frac{30}{100} \times 120 = 36$ gm

in second case weight of solute $= \frac{70}{100} \times 160 = 112$ gm

weight of solvent = (120 + 160) – (36 + 112) = 132

molality $= \frac{148/100}{132/1000} = 11.21$

27)

Let assume 34Cl has x gram and 38Cl has
(7– x) gram
moles of ^{34}Cl = x/34 and moles of
^{38}Cl = (7 – x) / 38

$M_{avg.} = \frac{7}{(x/34) + (7-x)/38}$

put the value of $M_{avg.}$ find the value of x,
so x = 5.1 g
moles of ^{34}Cl = 5.1/34 = 0.15
and moles of ^{38}Cl = (7 – 5.1)/38 =.05
n = 0.15 × 17 + 0.05 × 21 = 3.6
p = 0.15 × 17 + 0.05 × 17 = 3.4
 p + n = 7

28)

	A^{+n}	$\rightarrow$	$A^{+(n+4)}$
t = 0	a		0
t = 10 min	(a – x)		x

Let n-factor of reducing agent = n
M × n × 30 = a × 2 ...(1) [M = Molarity of reducing agent]
M × n × 45 = (a – x) × 2 + x × 5 ...(2)
M × n × 45 = M × n × 30 + 3x
M × n × 15 = M × n × 30 + 3x
M × n × 15 = 3x ⇒ x = M × n × 5
a = M × n × 15

$k = \frac{1}{10} \ell n \left(\frac{M \times n \times 15}{M \times n \times 10} \right) = 0.04 \text{ min}^{-1}$

29) Initial moles of H_2SO_4

$= 1.3 \times 3500 \times \frac{40}{100} \times \frac{1}{98} = 18.571$

Final moles of $H_2SO_4 = 1.2 \times 3500 \times \frac{20}{100} \times \frac{1}{98} = 8.571$

⇒ moles of H_2SO_4 consumed = 10

 It is evident from the given cell reaction that per mole of H_2SO_4 consumed involves transfer of two moles of electrons, total electron transferred = 20.

⇒ Q(withdrawn) = 20 × 96500 C

Ampere-hour = Q/3600 = 536.11

30)

X = PCl_3F_2, $P(CH_3)_3(CF_3)_2$, BF_3, CH_4 4

Y = $BF_{3....}$(1)

31) $_{26}Fe = (1s^2)\ (2s^2\ 2p^6)\ (3s^2\ 3p^6)\ (3d^6)\ (4s^2)$

σ = (2 × 1) + (8 × 0.85) + (7 × 0.35)

= 11.25

Hence, Z_{eff} = 26 – 11.25

x = 14.75

$\left(\frac{x}{10}\right) = 1.475$

Ans. 1.47 or 1.48

32) $E = eV_0 = \frac{hc}{\lambda}$

Given, $V_0 = 2 \times 10^4 V$

$$1.6 \times 10^{-19} \times 2 \times 10^4 = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$\lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{3.2 \times 10^{-15}} = 6.21 \times 10^{-11} m$$

= 0.621Å

PART 3 : MATHEMATICS

33) VA = 0 (Null matrix)

⇒ (loga)x + (logb)y + (logc)z = 0

(logb)x + (logc)y + (loga)z = 0

(logc)x + (loga)y + (logb)z = 0

which is homogenesis system of linear equations in x, y and z.

 x, y, z ≠ 0

⇒ system posseses non trivial solutions.

⇒ $$D = \begin{vmatrix} \log a & \log b & \log c \\ \log b & \log c & \log a \\ \log c & \log a & \log b \end{vmatrix} = 0$$

⇒ loga + logb + logc = 0

OR loga = logb = logc

 a, b, c are distinct

⇒ loga + logb + logc = 0

⇒ abc = 1

⇒ $b^3 = 1$ ($\because b^2 = ac$)

⇒ $b = 1 \Rightarrow ac = 1 \Rightarrow c = \frac{1}{a}$

⇒ System become

$(\log a)x - (\log a)z = 0$

$-(\log a)y + (\log a)z = 0$

$-(\log a)x + (\log a)y = 0$

⇒ $x - z = 0$

$y - z = 0$

$x - y = 0$

⇒ $x = y = z$

⇒ solutions lie on $\frac{x}{\alpha} = \frac{y}{\beta} = \frac{z}{\gamma}$

⇒ $\alpha : \beta : \gamma = 1 : 1 : 1$

$\frac{\alpha}{\beta} + \frac{\beta}{\gamma} + \frac{\gamma}{\alpha} = 3$

34) A_1 is cofactor matrix of A

$\Rightarrow A_1 = (\text{Adj}(A))'$

$\Rightarrow |A_1| = |\text{Adj}(A)|$

A_2 is cofactor matrix of A_1

$\Rightarrow |A_2| = |\text{Adj}(A_1)| = |\text{Adj}(\text{Adj}(A))|$

$$\Rightarrow |A_n| = \left|\frac{\text{Adj}(\text{Adj}(\ldots\ldots\ldots(\text{Adj}(A)))\ldots\ldots\ldots\ldots)}{\text{n times}}\right| = \lambda$$

$$= |A|^{(n-1)^n} \Rightarrow |A| = \lambda^{\frac{1}{(n-1)^n}}$$

35)
$$D_2 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix}$$

$$D_1 = \begin{vmatrix} ab^2 - ac^2 & bc^2 - a^2b & a^2c - b^2c \\ ac - ab & ab - bc & bc - ac \\ c - b & a - c & b - a \end{vmatrix}$$

$$D_1 = \begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{vmatrix}$$

$$D_1 D_2 = \begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{vmatrix} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= \begin{vmatrix} D_2 & 0 & 0 \\ 0 & D_2 & 0 \\ 0 & 0 & D_2 \end{vmatrix}$$

$D_1\ D_2 = D_2^3$

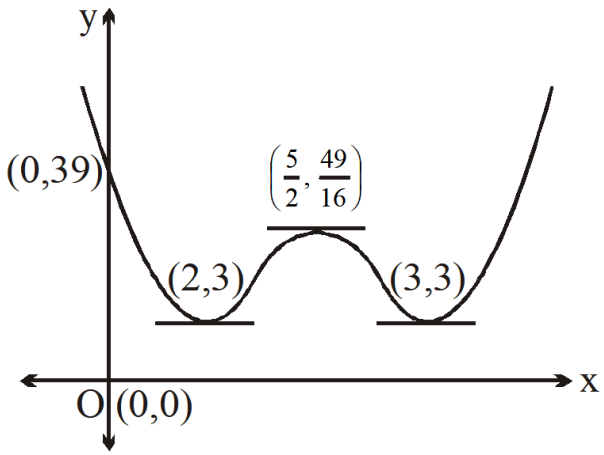
36)

$h'(x) = f(x)$

$$\int h(x)dx = g(x)$$

$h(x) = g'(x)$

$h'(x) = g''(x) = f(x)$



$$\int_{-\pi/2}^{\pi/2} (g(x) + g''(x)) \cos x\, dx = 5$$

$\therefore$ Apply IBP

$$\int_{-\pi/2}^{\pi/2} g(x)\ \cos x\, dx + \int_{-\pi/2}^{\pi/2} g''(x)\ \cos x\, dx = 5$$

$$\Rightarrow \int_{-\pi/2}^{\pi/2} g(x)\ \cos x\, dx + (\cos x\, g'(x))_{-\pi/2}^{\pi/2} + \int_{-\pi/2}^{\pi/2} \sin x\, g'(x)\, dx = 5$$

$$\Rightarrow \int_{-\pi/2}^{\pi/2} g(x)\ \cos x\, dx + (\sin x\, g(x))_{-\pi/2}^{\pi/2} - \int_{-\pi/2}^{\pi/2} \cos x\ g(x)\, dx = 5$$

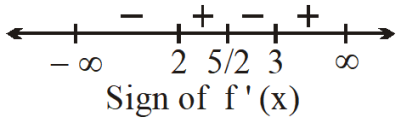
$$g\left(\frac{\pi}{2}\right) + g\left(-\frac{\pi}{2}\right) = 5 \Rightarrow 3 + g\left(-\frac{\pi}{2}\right) = 5 \Rightarrow g\left(\frac{-\pi}{2}\right) = 2$$

Let $p(x) = k(x-2)^2 (x-3)^2 + 3$

As, $p(1) = 4k + 3 = 7$

$\therefore$ $k = 1 \Rightarrow f(x) = (x-2)^2 (x-3)^2 + 3$

Now, $p'(x) = 2(x-2)(x-3)(2x-5)$



Sign of f ' (x)

$\Rightarrow$ p (x) has local maximum at $x = \frac{5}{2}$

$\therefore$ $p_{max.}\left(x = \frac{5}{2}\right) = \frac{49}{16}$

37) First arrange 14 black balls, 14 white balls to be placed in 15 gaps (c_1, c_2, c_3, ... c_{15}) such that $c_1+c_2+ c_3+ ....+ c_{15} = 14$ $(0 \leqslant c_i \leqslant 6)$

coefficient of x^{14} in $(1+x+x^2+...+x^6)^{15} = {}^{28}c_{14} - 15.{}^{21}c_7 + {}^{15}c_2$

38) $f(x) = \left(x^3 + \frac{1}{x^3}\right) - 6\left(x^2 + \frac{1}{x^2}\right) + 12\left(x + \frac{1}{x}\right) + a \geqslant 0$ $\forall\ x > 0$ and

$f(x) = \left(x^3 + \frac{1}{x^3}\right) - 6\left(x^2 + \frac{1}{x^2}\right) + 12\left(x + \frac{1}{x}\right) + a \leqslant 0$ $\forall\ x < 0$

Let $x + \frac{1}{x} = t$

Now

$(t)(t^2 - 3) - 6(t^2 - 2) + 12(t) + a \geq 0$ for ≥ 2

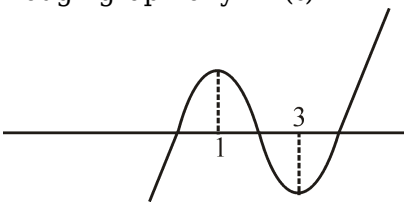
& $(t)(t^2 - 3) - 6(t^2 - 2) + 12t + a \leq 0$ for ≤ 2

Let $f(t) = t^3 - 6t^2 + 9t + 12 + a$

$f'(t) = 3t^2 - 12t + 9$

$= 3(t - 3)(t - 1)$

Rough graph of y = f(t)



for $t \geq 2$, $f(t) \geq 0 \Rightarrow f(3) \geq 0$

$\Rightarrow \boxed{a \geqslant -12}$

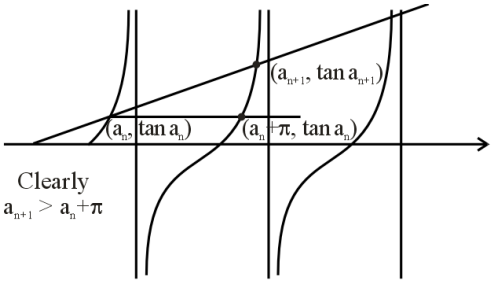
$f(-2) \leq 0$

$-8 - 6(4) - 18 + a + 12 \leq 0$

$a - 38 \leq 0$

$\boxed{a \leqslant 38}$

$a \in [-12, 38]$



39)

$\frac{dy}{dx} = \cos x\Big|_{x=a_n} = \frac{\sin a_n}{a_n + \frac{\pi}{2}}$

$a_n + \frac{\pi}{2} = \tan a_n$

Taking intersection of y = tanx

and $y = x + \frac{\pi}{2}$

40) $f(x) = \underbrace{\tan \pi x^3}_{u}\ \underbrace{\sqrt{9 + \tan^2 \pi x}\sqrt[3]{27 + \tan^3 \pi x}}_{v}$

$\Rightarrow f'(x) = u'v + uv'$

$u = 0$ at $x = -2$

$\Rightarrow f'(-2) = \sec^2\left(\pi x^3\right) 3\pi x^2 \sqrt{9 + \tan^2 \pi x}\sqrt[3]{27 + \tan^3 \pi x}$ at $x = -2$

$\therefore f'(-2) = 12\pi\ 3.3 = 108\pi$

$$g(x) = \frac{\cos^{-1}x\,(1-\cos x) + \tan^{-1}x\,(1-\cos x)}{(1+x^2)\cot^{-1}x}$$

$$= \frac{(1-\cos x)\left(\cos^{-1}x + \tan^{-1}x\right)}{(1+x^2)\cot^{-1}x}$$

g(x) = m.n

$\Rightarrow g''(x) = m"n + 2m'n' + mn"$; m = 1 – cos x

at x = 0; m = 0, m' = 0

$$\Rightarrow g''(0) = \cos x \cdot \left.\frac{\cos^{-1}x + \tan^{-1}x}{(1+x^2)\cot^{-1}x}\right|_{x=0}$$

$$= 1 \times \frac{\frac{\pi}{2}+0}{1\left(\frac{\pi}{2}\right)} = 1$$

41) $(\sin x + \cos x)(\tan^2 x - (1+\sqrt{3})\tan x + 3 + \sqrt{3}) = 3|\sin x + \cos x|$

Case I :

sinx + cosx ≥ 0

$(s+c)(t^2 - (1+\sqrt{3})t + \sqrt{3}) = 0$

$\Rightarrow t = -1, 1, \sqrt{3}$

$\Rightarrow x = \frac{3\pi}{4}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{7\pi}{4}$

Case II :

s + c < 0

$(s+c)(t^2 - (1+\sqrt{3})t + 6 + \sqrt{3}) = 0$

$\Rightarrow t = -1$ No solution

$\Rightarrow x = \frac{3\pi}{4}, \frac{7\pi}{4}$

$\Rightarrow$ sum of solutions is $\frac{37\pi}{12}$

$|p-q| = 25$

42) Let $a_k = \frac{3^k.\pi}{3^n+1}$ $1 + 2\cos(2a_k) = 1 + 2(1 - 2\sin^2 a_k)$

$= 3 - 4\sin^2 a_k$

$$= \frac{\sin 3a_k}{\sin a_k}$$

$$\prod_{k=1}^{2020} \frac{\sin 3a_k}{\sin a_k} = \frac{\sin\left(\frac{3^{k+1}\pi}{3^{2020}+1}\right)}{\sin\left(\frac{3^k.\pi}{3^{2020}+1}\right)}$$

$$= \frac{\sin\left(\frac{\pi.3^{2021}}{3^{2020}+1}\right)}{\sin\left(\frac{3\pi}{3^{2020}+1}\right)} = \frac{\sin\left(\frac{3\pi\left(3^{2020}+1-1\right)}{3^{2020}+1}\right)}{\sin\left(\frac{3\pi}{3^{2020}+1}\right)}$$

$$= \frac{\sin\left(3\pi - \frac{3\pi}{3^{2020}+1}\right)}{\sin\left(\frac{3\pi}{3^{2020}+1}\right)} = 1$$

43) $f(x + y) = f(x)f(y) - g(x)g(y)$ ...(i)
put $x = y = 0$ in (i)
$a = a^2 - b^2$ ($f(0) = a$, $g(0) = b$) ...(iii)
Also, $g(x + y) = g(x)f(y) + f(x)g(y)$ ...(ii)
put $x = y = 0$ in (ii)
$b = 2ab$ ...(iv)
from (iii) & (iv), we get only valid pair as $a = 1$, $b = 0$
i.e. $f(0) = 1$ & $g(0) = 0$
Now, $f'(x) = \lim_{y\to 0} \frac{f(x+y)-f(x)}{y}$

$$= \lim_{y\to 0} f(x)\left(\frac{f(y)-f(0)}{y}\right) - g(x)\left(\frac{g(y)-g(0)}{y}\right)$$

$\Rightarrow f'(x) = f'(0)f(x) - g'(0)g(x)$
$\Rightarrow f'(x) = -g'(0)g(x)$ (as $f'(0) = 0$) ...(v)
Similarly, from the second given relation
$g'(x) = f'(0)g(x) + g'(0)f(x)$
$\Rightarrow g'(x) = g'(0)f(x)$ (as $f'(0) = 0$) ...(vi)
Note $g'(0) \neq 0$, otherwise both f(x) & g(x) becomes constant
Now, (v) ÷ (vi)
$\frac{f'(x)}{g'(x)} = -\frac{g(x)}{f(x)} \Rightarrow f(x)f'(x) = -g(x)g'(x)$
$\Rightarrow 2f(x)f'(x) + 2g(x)g'(x) = 0$
Integrating :- $f^2(x) + g^2(x) = c = 1$
(as $f(0) = 1$, $g(0) = 0$)
Now, $\frac{1-(f^6+g^6)}{f^2g^2} = \frac{1-(f^2+g^2)(f^4+g^4-f^2g^2)}{f^2g^2}$

$= \frac{1-(1-3f^2g^2)}{f^2g^2} = 3$ (as $f^2 + g^2 = 1$)

44) $S = \sum_{k=1}^{n} (-1)^{k-1}\, {}^nC_k \left(1+\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{k}\right)$

$$= \sum_{k=1}^{n}\left((-1)^{k-1}\, {}^nC_k \sum_{r=1}^{k}\int_0^1 x^{r-1}dx\right)$$

$$= \sum_{k=1}^{n}(-1)^{k-1}\, {}^nC_k \int_0^1 \frac{1-x^k}{1-x}dx$$

$$= -\int_0^1 \sum_{k=0}^{n}(-1)^k\, {}^nC_k \frac{(1-x^k)}{(1-x)}dx$$

$$= -\int_0^1 -\frac{(1-x)^n}{1-x}dx$$

$$= \int_0^1 (1-x)^{n-1}dx = \int_0^1 x^{n-1}dx = \frac{1}{n}$$

45)

Given $A_{51} - A_{50} = 1000 \Rightarrow \ell_{51} w_{51} - \ell_{50} w_{50} = 1000$

$\Rightarrow (\ell_1 + 50d_1)(w_1 + 50d_2) - (\ell_1 + 49d_1)(w_1 + 49d_2) = 1000$

$\Rightarrow (\ell_1 d_2 + w_1 d_1) = 10$ .. (1)

(As, $d_1 d_2 = 10$)

$\Rightarrow$ (As, $d_1 d_2 = 10$)

$A_{100} - A_{90} = (\ell_1 + 99d_1)(w_1 + 99d_2) - (\ell_1 + 89d_1)(w_1 + 89d_2)$

$= 10(w_1 d_1 + \ell_1 d_2) + 188 \times 10 \times 10$

$= 100\ (1 + 188) = 100(189)$

$= 18900$

46) $T_k = \cos^{-1}\left(\frac{1}{k} \cdot \frac{1}{(k+1)} + \frac{\sqrt{(k-1)k(k+1)(k+2)}}{k(k+1)}\right)$

Let $x = \frac{1}{k}$ and $y = \frac{1}{k+1}$ [$\cos^{-1}x - \cos y = \cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$]

$\sqrt{1-x^2} = \sqrt{1-\frac{1}{k^2}} = \frac{\sqrt{k^2-1}}{k} = \frac{\sqrt{(k-1)(k+1)}}{k}$

$\sqrt{1-y^2} = \sqrt{1-\frac{1}{(k+1)^2}} = \frac{\sqrt{(k+1)^2-1}}{k+1} = \frac{\sqrt{k(k+2)}}{k+1}$

T_k is in form of $\cos^{-1}\left(xy + \sqrt{1-x^2} \cdot \sqrt{1-y^2}\right) = \cos^{-1}(y) - \cos^{-1}(x)$ ($y < x$)

$T_k = \cos^{-1}\left(\frac{1}{k+1}\right) - \cos^{-1}\left(\frac{1}{k}\right)$, substituting n = 2, 3, 4 .....

$T_2 = \cos^{-1}\left(\frac{1}{3}\right) - \cos^{-1}\left(\frac{1}{2}\right)$

$T_3 = \cos^{-1}\left(\frac{1}{4}\right) - \cos^{-1}\left(\frac{1}{3}\right)$

⋮ ⋮ ⋮

$T_n = \cos^{-1}\left(\frac{1}{n+1}\right) - \cos^{-1}\left(\frac{1}{n}\right)$

————————————

Sum $= \lim_{n\to\infty} \cos^{-1}\left(\frac{1}{n+1}\right) - \cos^{-1}\left(\frac{1}{2}\right) = \cos^{-1}(0) - \cos^{-1}\left(\frac{1}{2}\right)$

Sum $= \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$

$\Rightarrow S_n = \frac{\pi}{6} = \frac{120\pi}{k}$

$\Rightarrow$ k = 720 Ans.

$$\alpha+\beta=\frac{-b_1}{a_1} \text{ and } \alpha\beta=\frac{c_1}{a_1}$$

$$\Rightarrow (1+\alpha)(1+\beta)=1+(\alpha+\beta)+\alpha\beta=1-\frac{b_1}{a_1}+\frac{c_1}{a_1}=\frac{a_1-b_1+c_1}{a_1}$$

$$\Rightarrow (1+\alpha)^2(1+\beta)^2(1+\gamma)^2=\prod_{i=1}^{3}\left(\frac{a_i-b_i+c_i}{a_i}\right)$$

$$\Rightarrow 1+\sum\alpha+\sum\alpha\beta+\alpha\beta\gamma=\left(\prod_{i=1}^{3}\frac{a_i-b_i+c_i}{a_i}\right)^{1/2}$$

47) $$\Rightarrow \sum\alpha+\sum\alpha\beta+\alpha\beta\gamma=\left(\prod_{i=1}^{3}\frac{a_i-b_i+c_i}{a_i}\right)^{1/2}-1 \Rightarrow k=-1.$$

48)

$\log_8(x+1)=0$
$\Rightarrow x+1=1 \Rightarrow x=0$
$\log_3(\log_5 y)=0$
$\Rightarrow \log_5 y=1 \Rightarrow y=5$
$\log_5(3-\log_2 z)=0$
$\Rightarrow 3-\log_2 z=1 \Rightarrow z=4$
$(x+yz)^{-100}=20^{-100}$
$N=20^{-100}$
$\log N=-100\log_{10}20$
$=-100\log_{10}(2\times 10)$
$=-100(0.301+1)$
$=-130.1$